

OXFORD

macroeconomics

INSTITUTIONS, INSTABILITY, AND INEQUALITY

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Module code

Chapter 5: Money, Banking and the Macro-economy

Objectives:

By the end of this chapter, students should understand the following:

- How to incorporate a lending rate and a policy rate in the 3 equation model, including how a central bank responds to a shock
- How a (simplified) financial system works, including the money market, the yield curve, and bonds
- What factors determine the markup a profit-maximizing bank applies over the policy rate
- Credit rationing and information asymmetries
- Fractional reserve banking: credit risk, liquidity risk and solvency risk
- Why bank balance sheets matter: the role of equity, and the role of deposits
- Monetary policy frameworks in both the scarce reserves regime and ample reserves regime

When Cheap Money Wasn't Enough

- Imagine it's **2008**. The U.S. economy is falling apart — Lehman Brothers has collapsed, stock markets are crashing.
- The **Federal Reserve** reacts fast: it **slashes interest rates** almost to zero. In theory, this should make borrowing cheaper, encourage spending, and revive demand.
- But here's the twist: **households and firms couldn't borrow, even at those low rates**. Why? Because **banks stopped trusting each other**. They thought: *"What if the borrower defaults? What if the other bank fails tomorrow?"*
- So banks **added a big markup** on top of the central bank's policy rate. Lending dried up. Families couldn't refinance mortgages, businesses couldn't roll over loans.
- This is why we called it a **credit crunch**: the central bank opened the tap, but almost no water flowed.

Model Link

- **IS curve**: Policy rate down, but effective lending rates didn't follow → IS didn't shift as expected.
- **PC curve**: With unemployment soaring, inflation pressures collapsed.
- **MR curve**: Central banks had to invent *new tools* (QE, forward guidance) because the usual channel broke.



Policy rate vs lending rate = wholesale vs retail price

- CB sets the *wholesale* price (policy rate).
- Banks add a *markup* depending on risk → final *retail* price is the lending rate.
- Analogy: like coffee beans → Starbucks sets a higher final cup price to cover costs, risk, and profit.



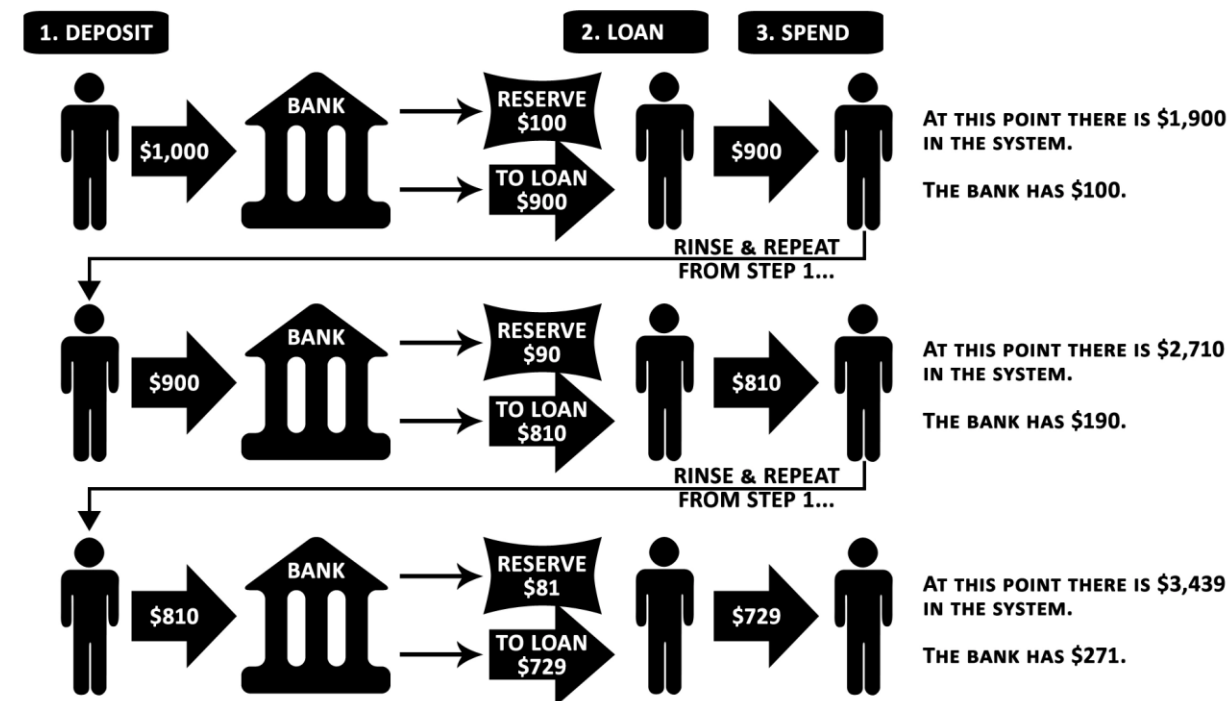
Banking markup grows in crises

- Even if CB cuts policy rates, banks may not pass them fully to borrowers.
- Example: 2008–09 financial crisis, when interbank lending dried up → banks widened markups due to fear of defaults.
- Markup components:
 - 1. Cost of Funds - the interest rate a bank pays to acquire the money it lends out, often from deposits or money markets.
 - Banks must pay interest on funds they collect from savers and earn a higher interest rate on the loans they provide to borrowers.
 - 2. Risk Premium - an additional amount charged to compensate the bank for the possibility that the borrower may not repay the loan in full or on time.
 - This premium is influenced by the borrower's credit score and the inherent risk associated with the loan.
 - 3. Administrative Costs - the expenses a bank incurs to manage the loan, including processing applications, handling payments, and covering overhead expenses like wages and rent.
 - A portion of these operational costs is factored into the final interest rate charged to the borrower.

Fractional reserve banking

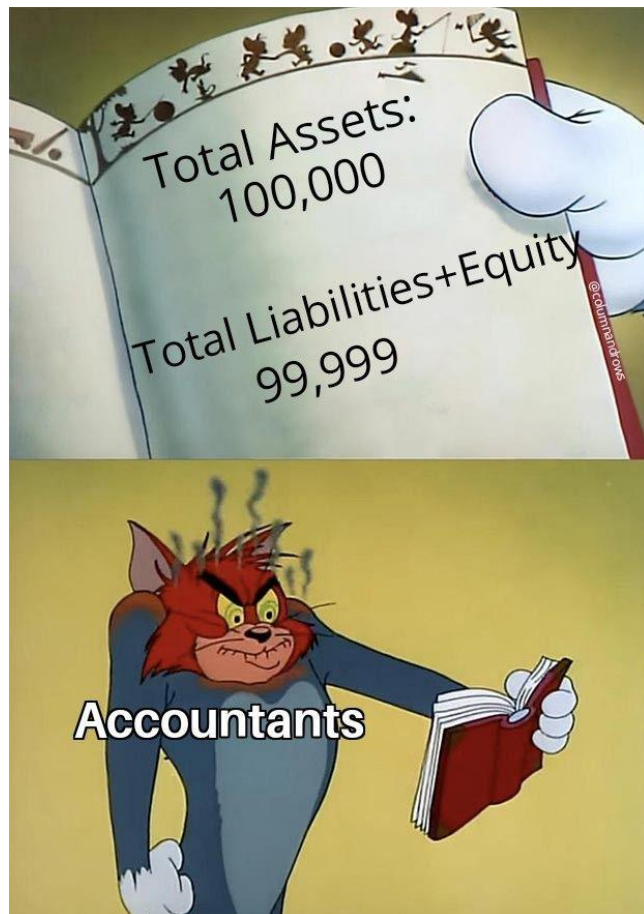
- Banks keep only a fraction of deposits liquid.
- *Analogy*: coat check at a party → if everyone comes at once to pick up their coats, chaos (bank run).
- *Real case*: Silicon Valley Bank 2023 run — withdrawals drained liquidity.

THE BASIC FRACTIONAL RESERVE BANKING CYCLE



[HTTP://CYNIC.ME](http://cynic.me)

Balance sheets matter



- Equity = cushion. High leverage = fragile.
- Example: Lehman Brothers 2008, Barclays (36x leverage pre-crisis).

Scarce vs ample reserves



- *Scarce*: pre-2008 — small adjustments in reserves controlled the policy rate.
- *Ample*: post-QE — CB floods system with reserves, so it steers policy via interest on reserves.
- *Analogy*: steering a bike (scarce reserves = delicate balance, ample = training wheels + heavy steering).

Introduction:

The banking system's primary function is to intermediate between borrowers and lenders

New elements to be introduced to the model:

- Distinction between the policy interest rate and the lending rate
- Commercial banks: set the lending rate, create bank money
- Money market: interbank market and market for government bonds
- Balance sheet of banks: solvency and liquidity problems, the role of equity
- Inflation-targeting central bank: scarce reserves regime and ample reserves regime

Interest rates: policy interest rate and lending interest rate

a. The 3-equation model and the banking markup

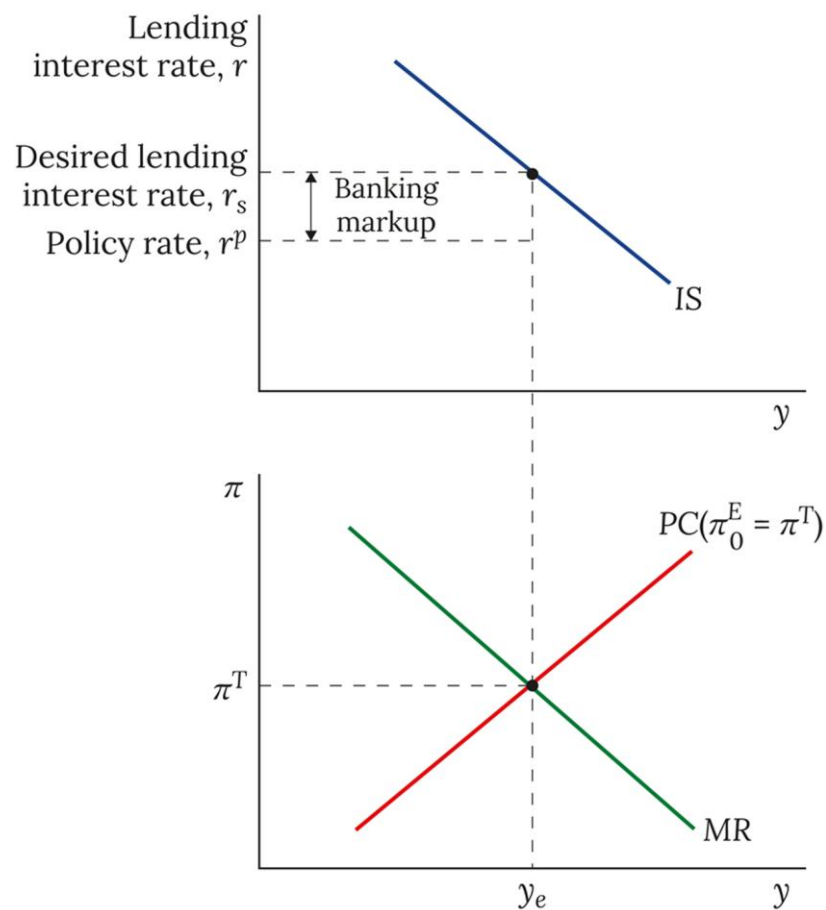


Figure 5.1(a) The 3-equation model and the banking markup.

The Central Bank (CB) policy rate (r^P) vs the lending rate (r)

- r is the rate relevant to the IS curve, and is a markup over r^P
- After the CB chooses the optimal output gap, it finds the desired lending interest rate r_s from the IS. Given its knowledge of the banking system (i.e., banking markup), it then sets r^P to achieve this
- Banking markup = $r - r^P$

Interest rates: policy interest rate and lending interest rate

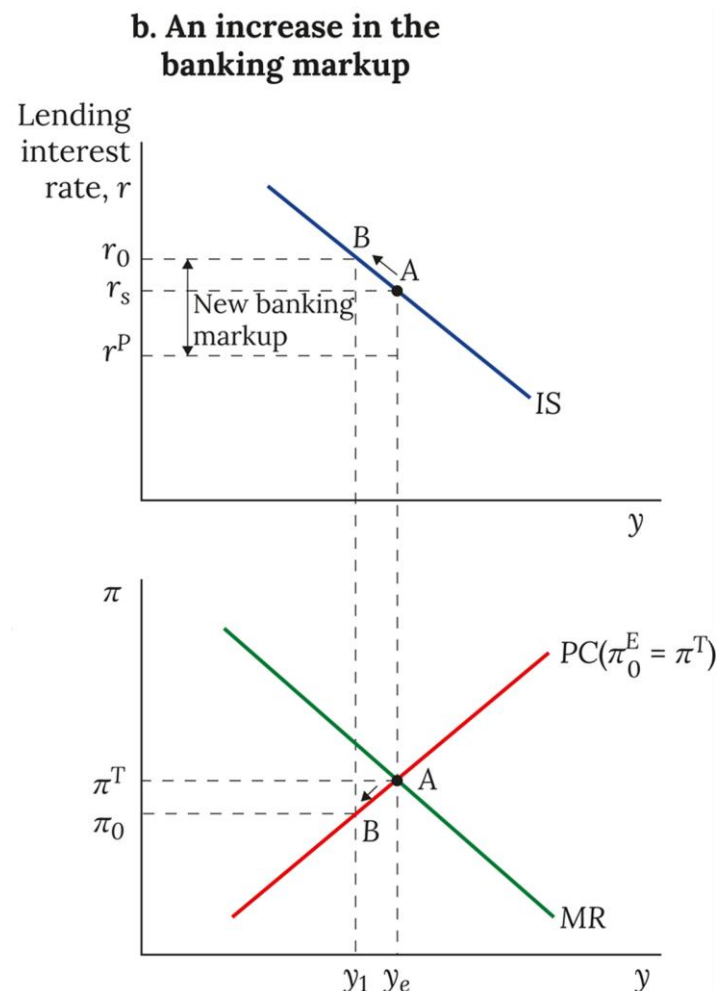


Figure 5.1(b) The 3-equation model and the banking markup.

A shock to the banking system:

- Banks view loans as being riskier $\rightarrow \uparrow$ markup
- Unanticipated by CB so r^P stays fixed $\rightarrow \downarrow$ AD
- Output falls to y_1 and inflation falls to π_0 (point B)

Under an inflation targeting CB, the adjustment process is as before (not pictured):

- CB finds desired output gap using intersection of $PC(\pi_1^E = \pi_0)$ and the MR
- CB finds lending rate on IS to generate this output
- CB lowers r^P to achieve this lending rate

Interest rates: three interest rates

In practice, there are many different interest rates, ranging from credit card rates to mortgage rates to the policy rate

To simplify, we consider only three interest rates:

- Lending rate: determines current output by the IS curve; set by market conditions
- Money market rate: determines cost of borrowing of the banks; set by market conditions
- Policy rate: directly controlled by central banks

Due to arbitrage, the money market rate and policy rate are assumed to be the same, and as such we only consider two rates in the model (policy and lending)

Interest rates: interest rates and the 3-equation model

The model		The UK in 2016 (annual average nominal rates)	
The lending rate	r	Rate on 5 year fixed rate mortgages	2.50
Capital and money market rate	Short-term interbank lending rates r^p	Sterling 2 week mean interbank lending rate	0.38
		Sterling overnight mean interbank lending rate	0.36
	Short- and medium-term government bond yields r^p	5 year government bond yield	0.65
The policy rate	r^p	Official Bank Rate	0.40

Figure 5.2 The assumptions made in the model about real interest rates and the associated nominal interest rates from the UK in 2016.
Source: Bank of England (data accessed September 2020). Unfortunately the relevant interbank lending rate series were discontinued at the end of 2016. Similar series are available, for example from the *Financial Times*.

Data on UK interest rates in 2016 is in line with assumptions of our model:

- Average short term money market rates (i.e., overnight and 2-week interbank rate) $\approx$ policy rate
- Lending rate represented by the mortgage rate $>$ the funding cost of banks

Interest rates: nominal and real interest rate

Throughout this chapter, we will mainly treat interest rates as the real interest rate:

- Lending rate: real interest rate
- Policy rate: nominal rate
 - Recall Fisher equation: $i \approx r + \pi^E$
 - With constant inflation expectations, changing i is the same as changing r
 - Hence, we can use a real rate r^P to represent the policy rate ($\Delta i = \Delta r$ if π^E is fixed.). We will relax this assumption when dealing with the Zero Lower Bound later in the Chapter
- Lending rate (e.g., what banks charge borrowers): treat it as the real rate r . → Because for households and firms, what matters is the inflation-adjusted cost of borrowing.
- Policy rate (e.g., the central bank's rate like Fed funds, ECB refi, BoK base rate): treat it as the nominal rate i . → Because the central bank directly sets a nominal number.

The financial system: key players and markets

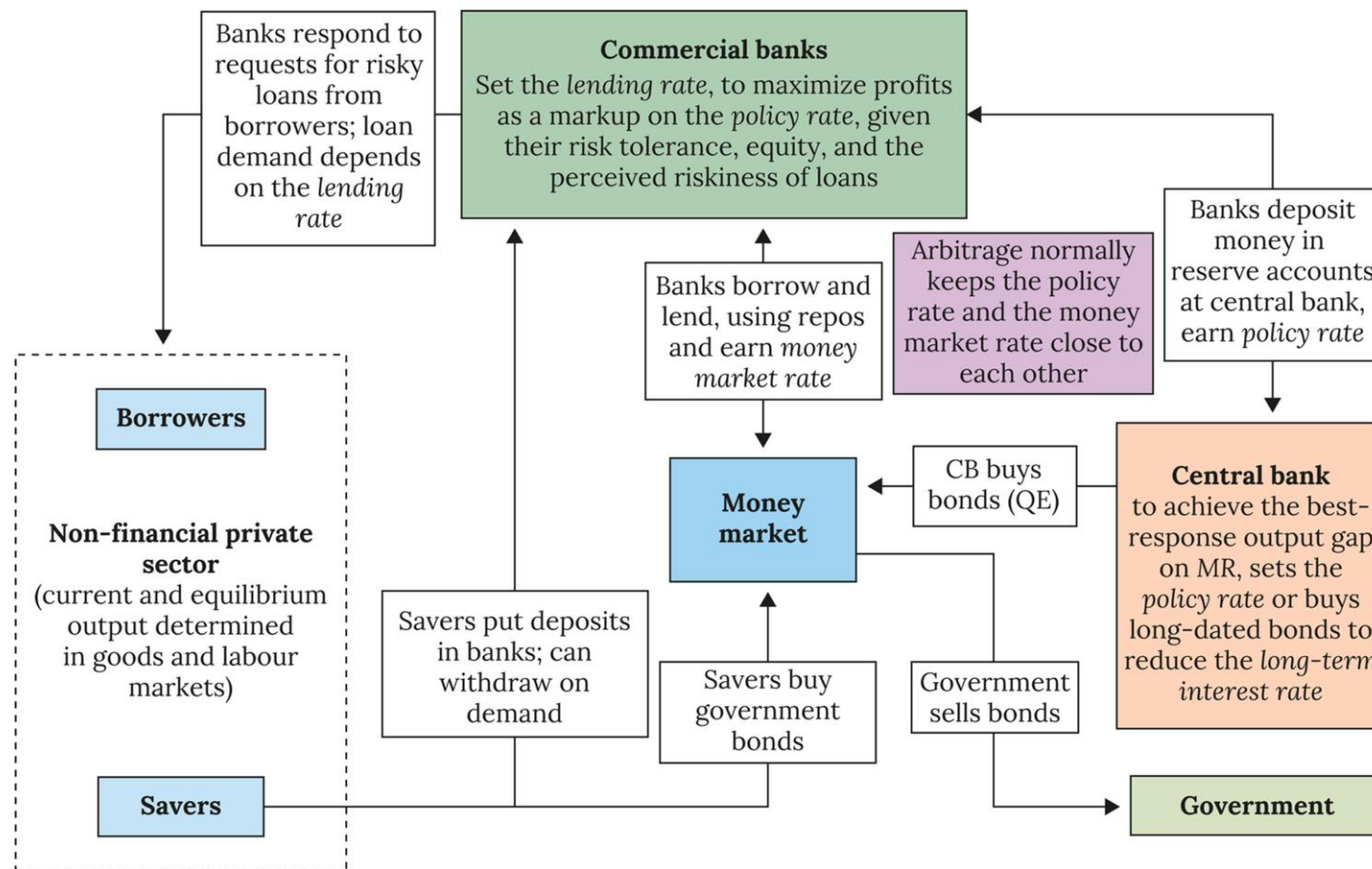


Figure 5.3 A simplified representation of the banking system.

The non-financial private sector:

- Comprised of households and firms; set current and equilibrium output; hence, together with the IS curve, determines stabilizing interest rate r_s

Commercial banks:

- Profit maximizing; lend to the private sector; create deposits; hold, borrow and lend reserves

The financial system: key players and market

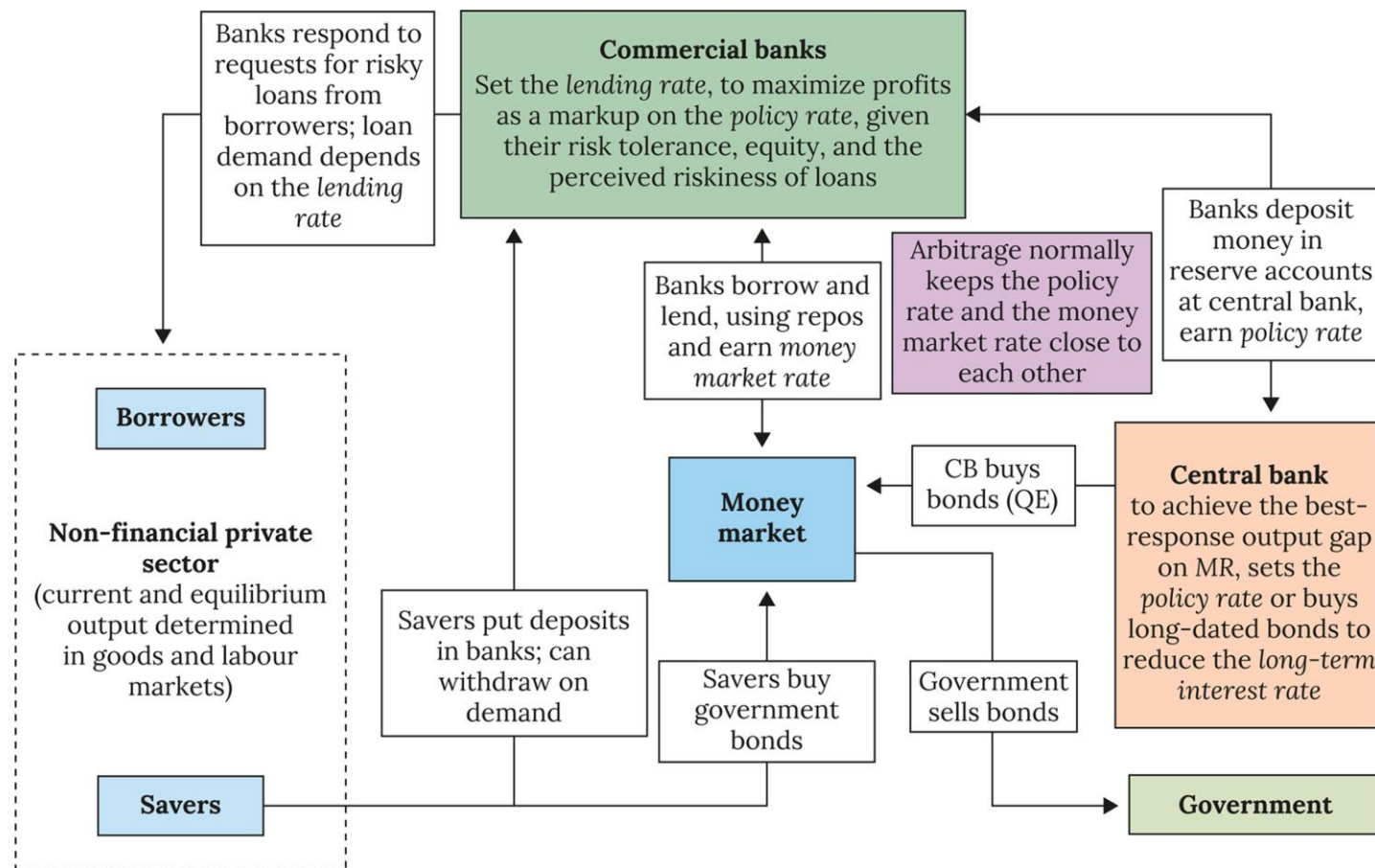


Figure 5.3 A simplified representation of the banking system.

The central bank:

- Keep commercial banks' reserves; set policy rate to achieve the desired lending rate (the rate which minimizes their loss function)

The money market :

- Where interbank transaction activities occur; any gap between the money market rate and the policy rate creates arbitrage opportunities

The financial system: collateral, bonds and repos

Collateral

- An asset that the lender will receive should the borrower default
- Secured borrowing: borrowing with collateral (e.g., housing market)

Bonds

- A financial instrument sold by an institution wishing to borrow to an investor wishing to lend
- Unlike bank loan, the initial buyer can sell the bond to another investor in the market

Repos

- Repurchase agreement: banks borrow by selling a bond in the repo and promising to buy it back the next day (or 14 days later) at a slightly higher price
- The main way that banks borrow and lend to each other in the money market

The yield curve: bonds

a. The yield curve

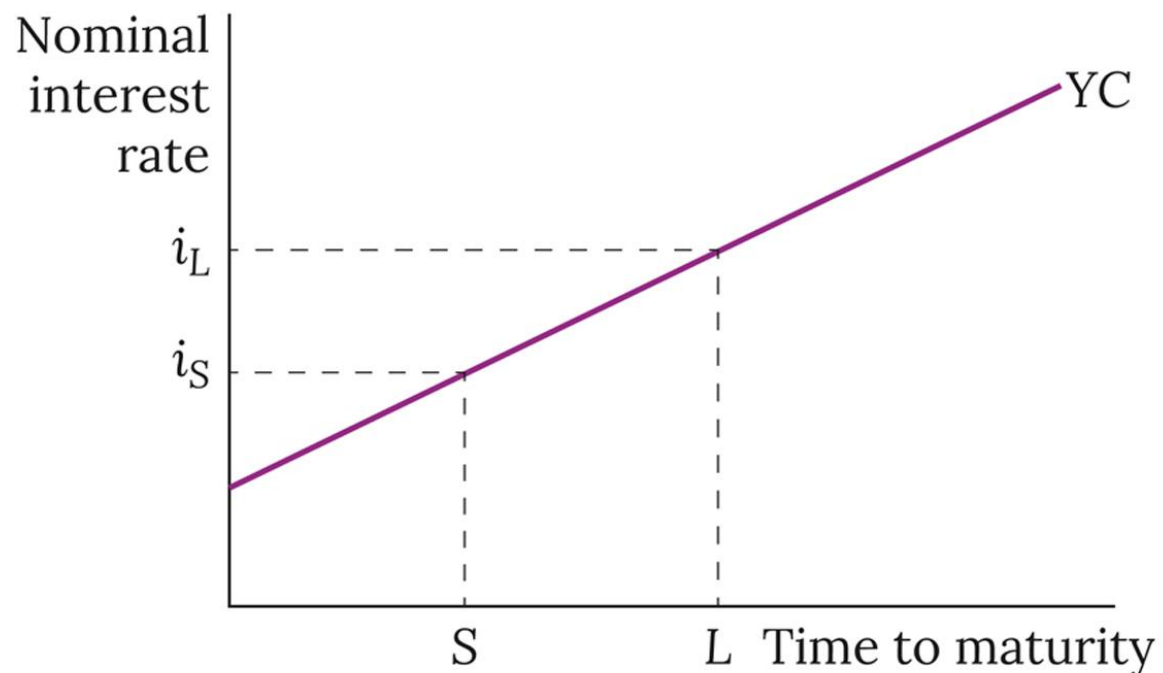


Figure 5.4(a) Monetary policy and the yield curve.

Note: In reality, the yield curve may not have a monotonic relationship with time to maturity. For instance if short-term rates are high but investors expect a recession in the near future, this suggests that interest rates will be cut soon to stimulate growth. Investors may therefore be happy to lock in a lower yield on long-term debt than on short-term debt, as they know that the relatively high short-term yields will not last long. This is known as 'yield curve inversion.'

The yield or interest rate of a bond changes inversely with its price:

$$i = (\bar{P} - P)/P$$

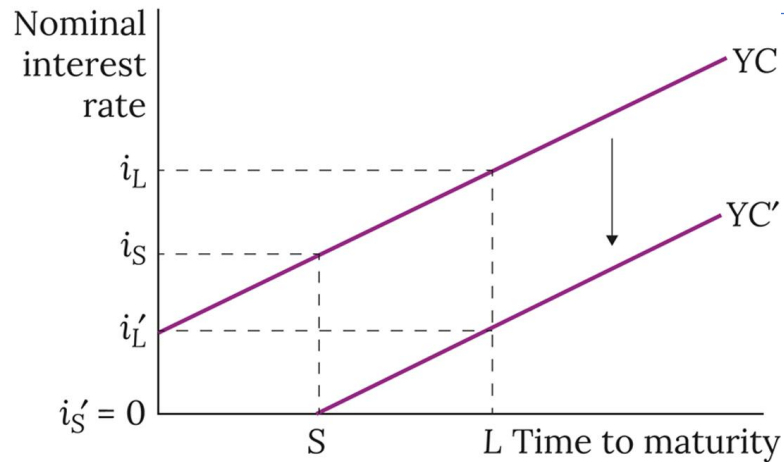
where $\bar{P}$ is the redemption value

Yet, yield increases with time to maturity, creating an upward sloping yield curve

- Investors demand a higher return to hold long-term bonds
- There is more uncertainty in the long term: future interest rate fluctuations; recessions in future decades

The yield curve: a negative demand shock

b. The yield curve at the zero lower bound



c. The yield curve and quantitative easing

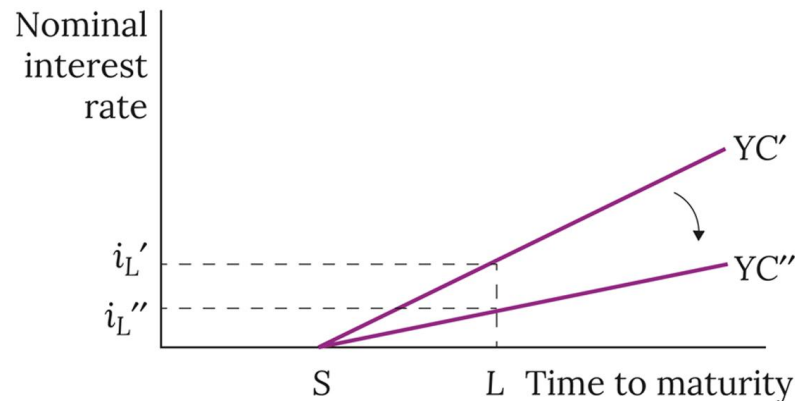


Figure 5.4(b and c) Monetary policy and the yield curve.

Note: In reality, the yield curve may not have a monotonic relationship with time to maturity. For instance if short-term rates are high but investors expect a recession in the near future, this suggests that interest rates will be cut soon to stimulate growth. Investors may therefore be happy to lock in a lower yield on long-term debt than on short-term debt, as they know that the relatively high short-term yields will not last long. This is known as 'yield curve inversion.'

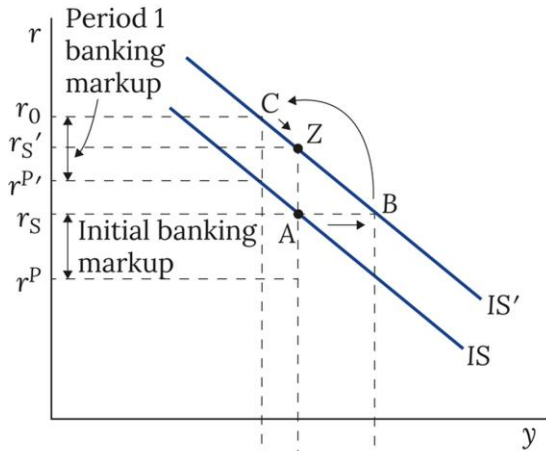
Under conventional monetary policy:

- $\downarrow$ policy rate by CB $\rightarrow \downarrow$ money market rate due to arbitrage $\rightarrow \downarrow$ lending rate by commercial banks
- As CB lowers policy rate to zero, the yield curve shifts down to YC' .
 - As the short-term nominal rate cannot fall below zero, the long-term rate cannot fall below i'_L
 - Given a large negative demand shock, this may be insufficient

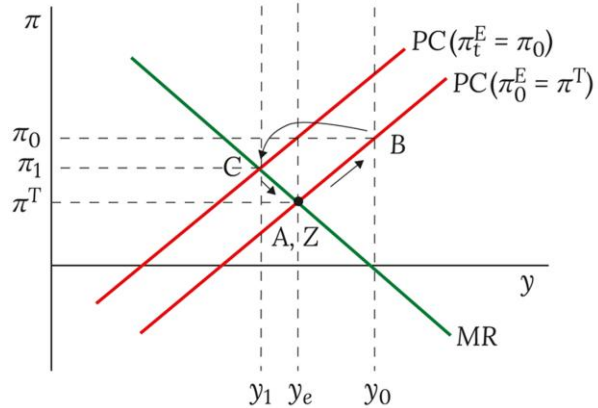
Under unconventional monetary policy:

- CB applies quantitative easing: purchase large quantities of long-term government bonds
- $\uparrow$ price of government bonds $\rightarrow \downarrow$ yield $\rightarrow$ the yield curve pivots, lowering long term nominal rates

Extended 3-equation model: investment boom



The increase in investment shifts the IS curve to the right. The central bank responds by raising the policy rate to $r^{P'}$ to achieve a lending rate of r_0 . Policy rate then lowered over following periods until the lending rate reaches r_S' .



The economy moves to B with rightward shift of IS:

- $\uparrow$ investment $\rightarrow \uparrow$ output and $\uparrow$ inflation

The CB chooses point C and targets r_0 :

- Given that CB cannot set lending rate directly, the CB uses its knowledge about markup and sets the policy rate as $r^{P'}$
 - $\uparrow$ policy rate by CB $\rightarrow \uparrow$ money market rate due to arbitrage $\rightarrow \uparrow$ lending rate by commercial banks $\rightarrow \uparrow$ cost of borrowing for firms and households $\rightarrow \downarrow$ investment $\rightarrow \downarrow$ output and $\downarrow$ inflation $\rightarrow$ the economy reaches C

In subsequent periods, the CB then gradually reduces the policy rate until the lending rate reaches its new stabilizing rate at r_S' , and inflation has returned to target

Figure 5.5 A permanent investment boom in the 3-equation model with the banking system. This is similar to the analysis done in Chapter 3, except that the central bank now sets the policy rate, targeting a desired lending rate.

The business of banking: money supply

Money creation:

- Base money: created by CB; banknotes + commercial banks' reserves
- Bank money: created by commercial banks when they make loans

Commercial banks use base money to cover withdrawals from customers (using banknotes held in vaults) and transactions with other banks (using reserves held at the central bank).

Money supply has various definitions:

- Typically, they don't include base money held by banks
- Narrow money: banknotes of the non-bank public + demand deposits
 - an indicator of spending in the economy
- Broad money also include term deposits
 - an indicator of future spending and of the transmission mechanism of monetary policy

The business of banking: money demand

Money demand can be given by the function:

$$\frac{M^D}{P} = f(y, i; \Phi)$$

- Increases with output (y): Income $\uparrow \rightarrow \uparrow$ transactions $\rightarrow$ hold more money
- Decreases with interest rate (i): $i \uparrow \rightarrow$ bonds more attractive $\rightarrow \uparrow$ opportunity cost of holding money $\rightarrow$ hold less money

Shifters of the demand for money function:

- Unanticipated inflation changes
- Structural factors in the financial sector (Φ):
 - Confidence; innovations in payment technology

The business of banking: money market equilibrium

Changes in the supply of money in the economy are the outcome of changes in the demand for money. These changes in money supply are entirely reactive, and do not themselves lead to further impacts on the wider economy; as will be seen, it is the demand and supply of reserves which has consequences on the wider economy.

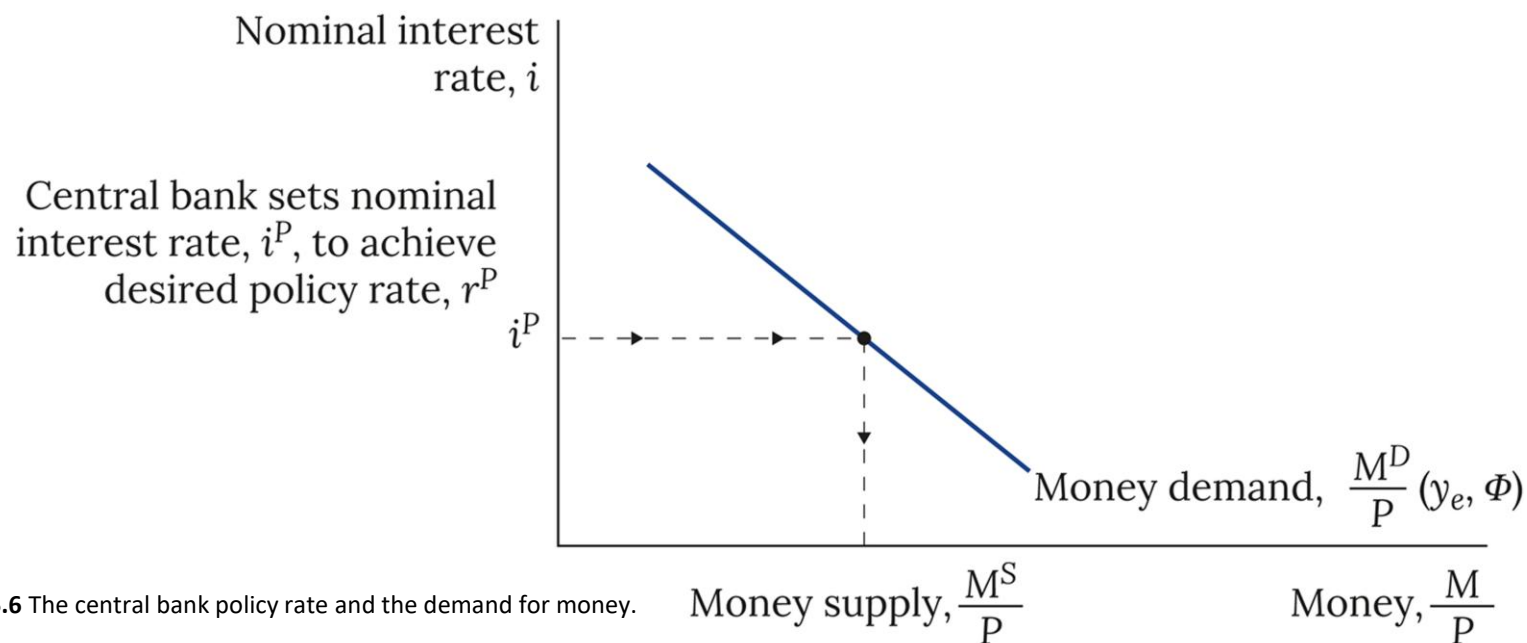


Figure 5.6 The central bank policy rate and the demand for money.

The business of banking: markup

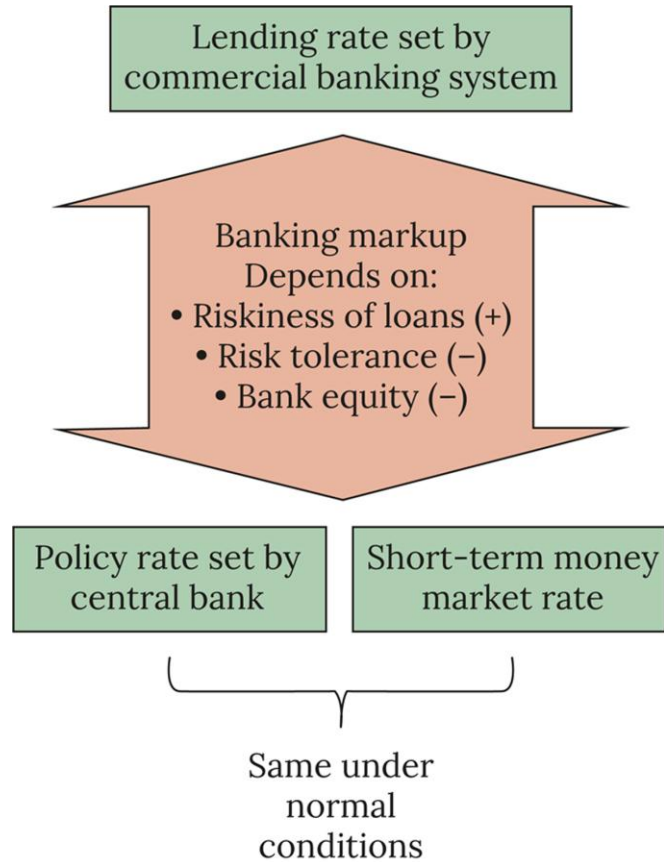


Figure 5.7 The relationship between the policy interest rate, the short-term money market rate, and the lending interest rate.

Banks choose the markup and set the lending rate based on:

- Riskiness of loans: $\uparrow$ credit risk $\rightarrow \uparrow$ markup
- Risk tolerance: $\downarrow$ risk tolerance $\rightarrow \uparrow$ markup for a given risk level
- Bank equity: $\downarrow$ equity/ capital cushion $\rightarrow \downarrow$ ability to bear credit risk $\rightarrow \uparrow$ markup
- If the banking sector is not competitive: $\uparrow$ market power $\rightarrow \uparrow$ markup

Lending rate: $r = (1 + \mu^B) r^P$

- markup (μ^B) increases with risk and decreases with risk tolerance and bank equity

The business of banking: markup

Stable relationship between r^P and r was disrupted by the financial crisis:

- Key transmission mechanism of monetary policy broke down in the post-crisis world

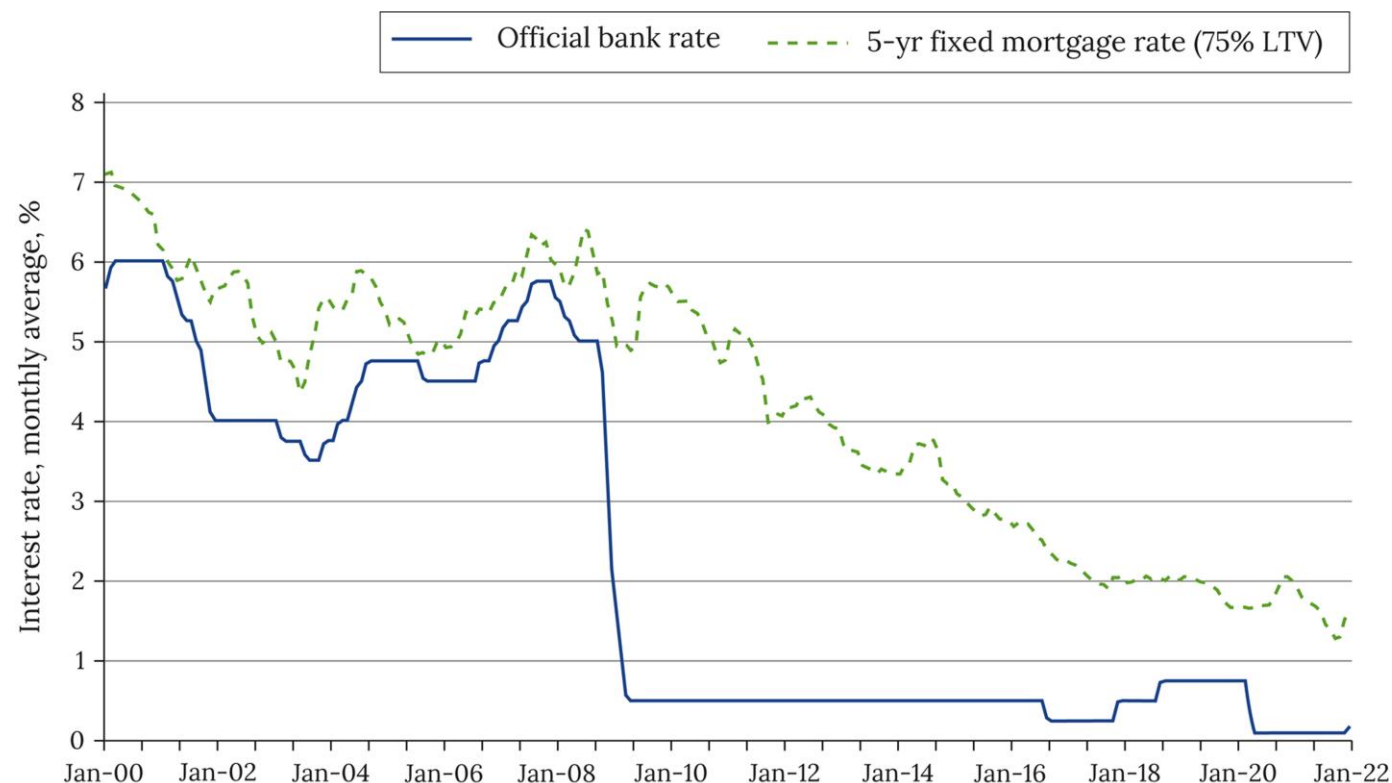


Figure 5.8 UK Official Bank Rates and 5-year fixed mortgage rates (75% Loan to Value ratio): Jan 2000–Dec 2021. A 75% Loan to Value mortgage is where the borrower has borrowed up to 75% of the value of the house, with the remainder coming from a deposit.
Source: Bank of England (data accessed June 2022).

The business of banking: credit rationing

Banks don't just adjust markups in response to credit risk, they also impose credit rationing

- Credit rationing affects household credit constraints → increases the multiplier and decreases the slope of the IS curve
- Information asymmetries contribute to credit rationing:
 - Moral hazard: Efforts exerted by borrowers in their projects are unobserved by the bank
 - Adverse selection: Individuals with stronger prospects will self-select away from high interest loan agreements → only weaker applicants remain → banks ration credit
 - E.g., Banks charge a higher interest rate as households' expected future earnings are unknown to them → households cannot borrow when needed

The business of banking: credit rationing

Given information problems, borrower's wealth (in terms of equity stakes or collateral) is important for signaling credibility to lenders and aligning incentives.

A borrower with a good project but no wealth will be denied credit, an outcome that is inefficient and unfair

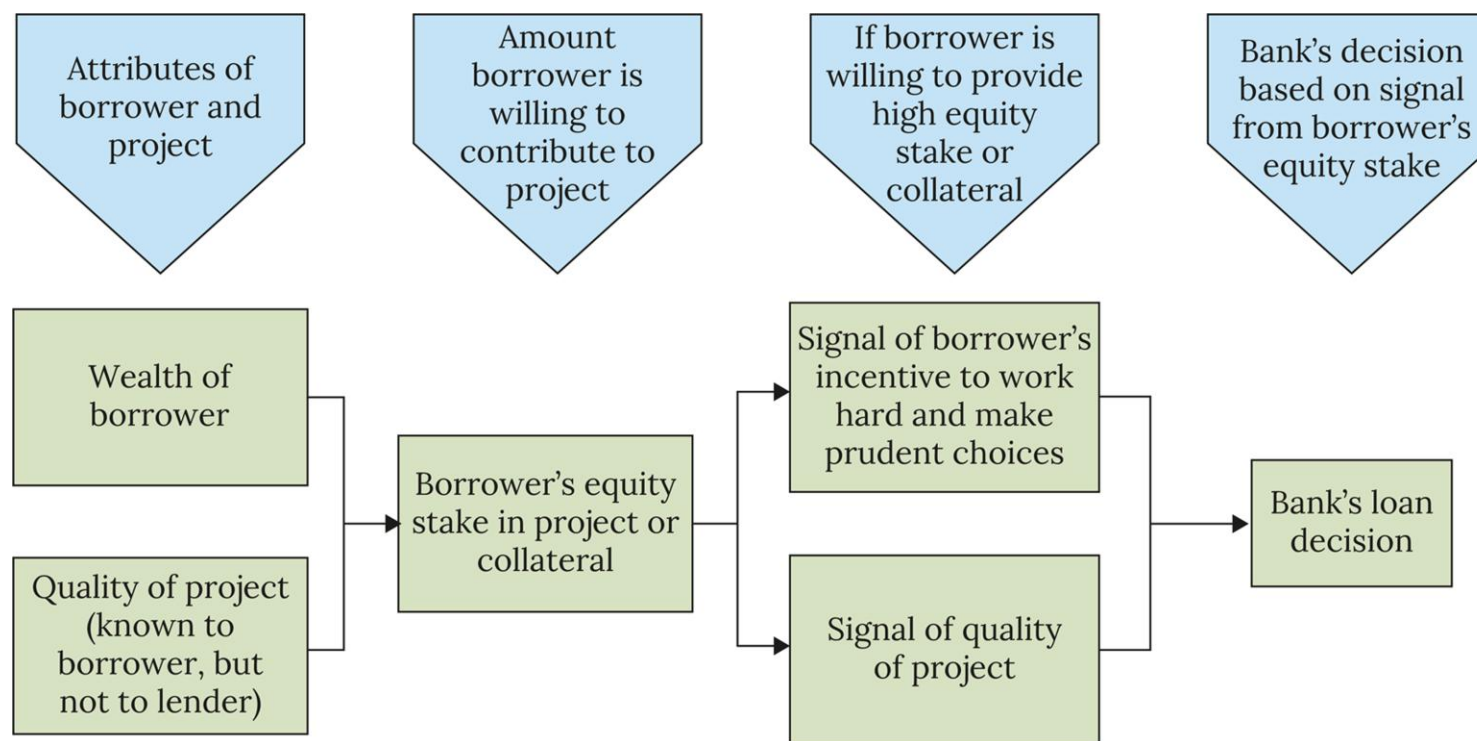


Figure 5.9 Credit constraints: the role of borrower wealth in the lending decisions of banks.

The business of banking: fractional reserve banking system

Fractional reserve banking system: Banks only hold a fraction of deposits in liquid form in their reserve accounts at the CB

The roles of banks:

- Maturity transformation: Transforming liquid savings into long-term borrowing, but a maturity mismatch poses liquidity risk to banks
- Aggregation: Aggregating small savings to make large loans
- Risk pooling: Banks can better withstand defaults → offer limited risk to savers

Liquidity risk: Inadequate reserves to meet depositors' demand to withdraw money from their accounts (e.g., “banking panic” → incentive to be the first to withdraw → everyone withdraws → bank run)

- The CB acts as a Lender of Last Resort (LOLR) to provide emergency liquidity, while the government insures deposits to reduce the likelihood of a bank run

The business of banking: fractional reserve banking system

Another danger the banks face is insolvency:

- Solvency risk: when the value of assets is less than that of debts or liabilities → bankruptcy if not bailed out by the government
- Interconnectedness in the banking sector:
 - “Fire sale”: an illiquid bank sells assets at discounted prices → asset prices drop → other banks face reductions in their assets’ values (e.g., housing) → they are forced to sell their assets
 - Solvency problem for a small number of banks → banks unsure about safety in lending to one another → widespread liquidity problems (‘credit crunch’)
- Insolvency has direct negative effects on bank depositors, creditors, shareholders and bondholders

Balance sheets: balance sheet identity

Assets	% of balance sheet	Liabilities	% of balance sheet
1. Banknotes and reserve account balances at the central bank	2	1. Deposits	50
2. Government bonds, which can be used as collateral for repo borrowing	10	2. Wholesale repo borrowing secured with collateral	30
3. Bonds and other financial assets issued by the private sector, some of which can be used as collateral for repo borrowing	20	3. Unsecured borrowing	16
4. Wholesale reverse repo lending	11		
5. Loans (e.g. mortgages)	55		
6. Fixed assets (e.g. buildings, equipment)	2		
Total assets	100	Total liabilities	96
		Net worth	
		Equity	4
Memorandum item: Leverage (Total assets/Net worth)		100/4 = 25	

Figure 5.10 The balance sheet of a typical commercial bank before the financial crisis.

Note: All items in the balance sheet are expressed as a percentage of total assets.

Balance sheets are useful to evaluate liquidity and solvency

Balance sheet identity:
 $\text{Net worth(equity)} \equiv \text{Assets} - \text{Liabilities}.$

If a bank has negative equity, it is insolvent

Balance sheets: asset side of the balance sheet

There are six main items on the asset side of the balance sheet (Figure 5.10):

1. Cash and reserves at the CB: highly liquid assets; low interest rate (r^P) so banks do not hold more than is prudent
2. Government bonds: interest-earning and low risk; can be used as high-quality collateral for borrowing from other banks
3. Asset-backed securities (ABS): securities backed by bundles of loans like mortgages; can be used as collateral for the bank's own borrowing if rated low risk
4. Wholesale reverse repo lending: see item 2 on "Liability Side"
5. Loans: the core activity of the bank; including mortgages, car loans etc.
6. Fixed Assets: Physical assets, e.g., buildings and equipment

Balance sheets: liability side of the balance sheet

There are three main items on the liability side of the balance sheet (Figure 5.10):

1. Deposits: made by households and firms; the largest liability of banks
2. Wholesale repo borrowing secured with collateral: As seen earlier, banks can sell an asset and agree to repurchase it shortly after at a higher price

For the borrowing bank, this is repo, a liability. For the lending bank, this is reverse repo, an asset.

Large banks perform a market maker function: lend and borrow at the same time in the repo market

3. Unsecured borrowing: banks can also borrow without collateral, e.g. reserves in the money market; lenders are not repaid if the borrowing bank becomes insolvent

Balance sheets: make loans – example

Banks seek to attract new deposits when making loans:

- Column 1: initially, Bank A has reserves that can cover 10% of its deposits
- Column 2: Bank A makes a £10 loan. Loan on the asset side and deposit on the liability side each increase by 10
- Column 3: the borrower spends the money at a business that uses Bank B. Bank A transfers £10 of deposits and reserves to Bank B. Now Bank A has no reserves and cannot meet liquidity requirements

	Column 1: initial situation	Column 2: £10 of new loans	Column 3: The £10 are spent at a different bank	Column 4: £10 of new deposits are attracted
Loans (asset)	100	110	110	110
Reserves (asset)	10	10	0	10
Deposits (liability)	100	110	100	110
Equity (liability)	10	10	10	10
Reserve to deposit ratio	10%	9.1%	0%	9.1%

Table 5.1 Why banks require deposits. Visit www.oup.com/he/carlin-soskice to explore Table 5.1 in more detail.
Source: Based on McLeay et al. (2014).

Balance sheets: make loans – example

Bank A has two options:

- Borrow reserves at the money market rate which can be expensive
- Attract new deposits from customers
 - Whenever new deposits come in, Bank A gains corresponding reserves as shown in Column 4

Therefore, Bank A can make loans without impacting its reserves even if its borrower spends the money elsewhere

	Column 1: initial situation	Column 2: £10 of new loans	Column 3: The £10 are spent at a different bank	Column 4: £10 of new deposits are attracted
Loans (asset)	100	110	110	110
Reserves (asset)	10	10	0	10
Deposits (liability)	100	110	100	110
Equity (liability)	10	10	10	10
Reserve to deposit ratio	10%	9.1%	0%	9.1%

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Source: Based on McLeay et al. (2014).

Balance sheets: leverage – Barclays example

Assets		Liabilities	
1. Banknotes and reserve account balances at the central bank	7,345	1. Deposits	336,316
2. Wholesale reverse repo lending	174,090	2. Wholesale repo borrowing secured with collateral	136,956
3. Loans (e.g. mortgages)	313,226	3. Unsecured borrowing	111,137
4. Fixed assets (e.g. buildings, equipment)	2,492	4. Trading portfolio liabilities	71,874
5. Trading portfolio assets, including government bonds	177,867	5. Derivative financial instruments	140,697
6. Derivative financial instruments	138,353	6. Other liabilities	172,417
7. Other assets	183,414		
Total assets	996,787	Total liabilities	969,397
		Net worth	
		Equity	27,390

Memorandum item: Leverage (Total assets/Net worth) $996,787/27,390 = 36.4$

Figure 5.11 The consolidated balance sheet of Barclays PLC in 2006, £m.
Source: Barclays PLC Annual Report 2006.

Another important information that balance sheets provide us is:

Leverage ratio = Assets / Equity

- A higher leverage ratio implies a smaller equity buffer and a greater insolvency risk

Example: Before the financial crisis, Barclays PLC has a leverage ratio of 36.4

Balance sheets: leverage – Barclays example

Barclays' leverage ratio continue to grow rapidly in the years leading up to the financial crisis, peaking in 2008; following the crisis Barclays de-leveraged by selling part of its loan portfolio (reducing risk) as well as issuing shares (raising equity)



Figure 5.12 The leverage ratio of Barclays PLC from 2000 to 2021 (defined as total assets/equity).

Source: Barclays PLC Annual Reports.

Balance sheets: central bank balance sheet

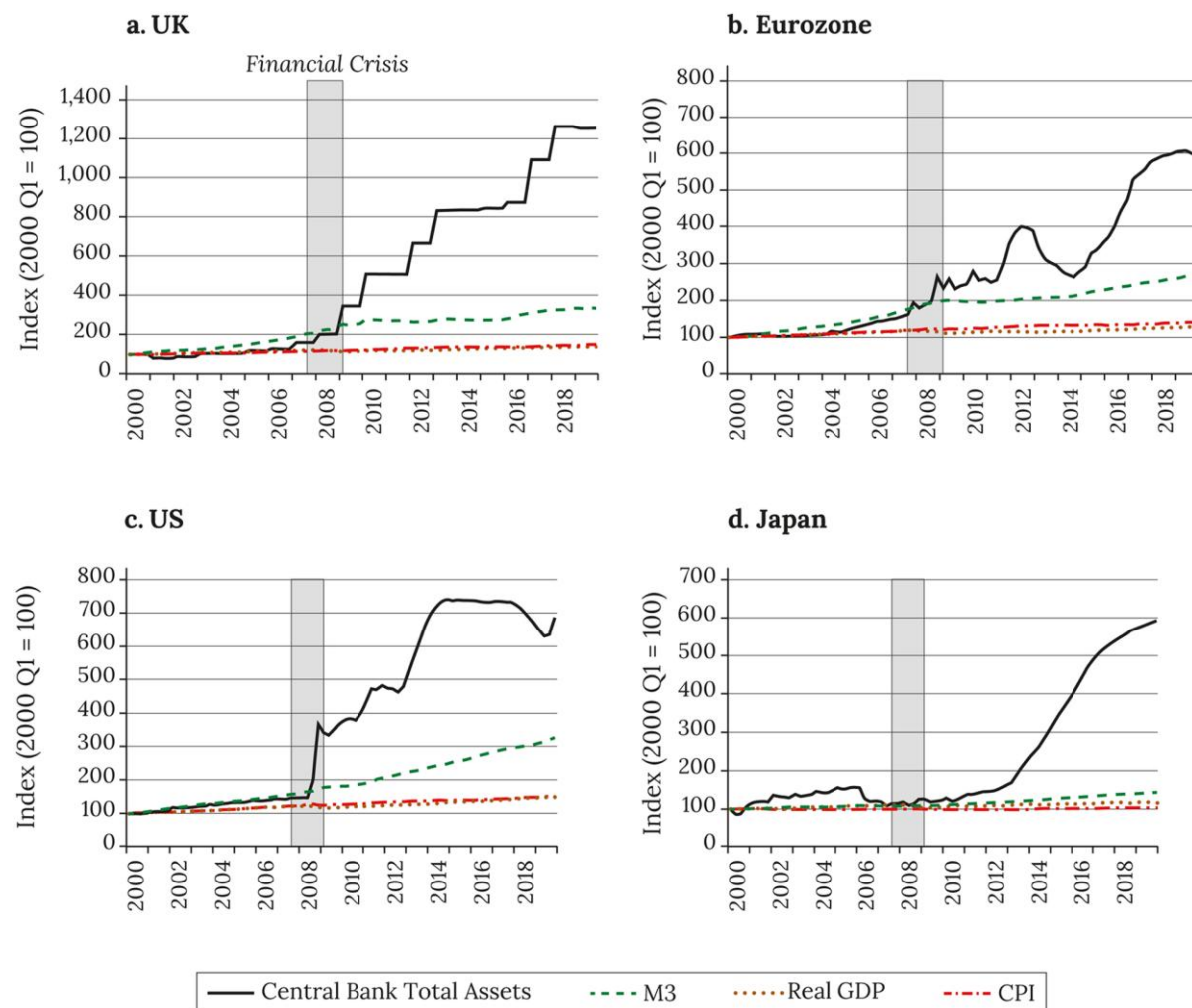


Figure 5.13 The total central bank assets, real GDP, M3 money supply and Consumer Price Index for the UK, the Eurozone, the US and Japan.
 Source: Federal Reserve Bank of St Louis, Bank of Japan, European Central Bank, Economic Statistics Centre of Excellence, and the OECD.

Central bank balance sheets have come to public attention under quantitative easing (QE) because the purchase of assets has grown rapidly:

- The growth in central bank assets has significantly outstripped growth in either real GDP, the consumer price index, or the money supply, suggesting that it has not been driven purely by economic fundamentals

Balance sheets: CB balance sheet

Central bank balance sheet also follows the balance sheet identity:

- $\text{Equity} \equiv \text{Assets} - \text{Liabilities}$
- Equity allows the central bank to absorb losses

Assets	% of balance sheet	Liabilities	% of balance sheet
1. Bond holdings	3	1. Reserve account balances	81
2. Open market operations, assets	2	2. Open market operations, liabilities	1
3. Foreign currency, reserves	1	3. Notes in circulation	14
4. Asset Purchase Facility (QE)	75	4. Foreign currency, liabilities	1
5. Other assets	19		
Total assets	100	Total liabilities	97
		Net worth	
		Equity	3

Figure 5.14 An example Central Bank balance sheet, loosely based on the Bank of England's balance sheet in January 2020. The Asset Purchase Facility is used by the Bank to implement quantitative easing (QE).

Balance sheets: CB balance sheet

Key item on liability side: reserve account balances

- Demand deposits of commercial banks at CB which help settle transactions
- The total quantity of reserves remain fixed by the CB

Key item on asset side: asset purchase facility

- Assets acquired through quantitative easing

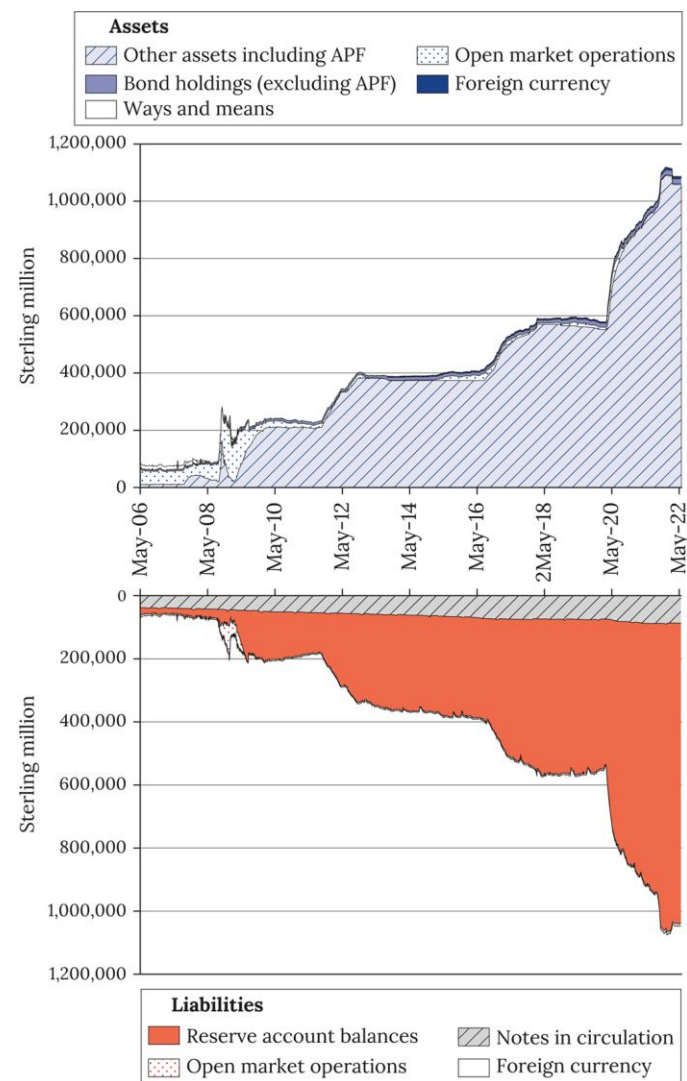
Difference between Open Market Operations (OMO) and quantitative easing (QE):

- OMO: used to adjust quantity of reserves to target an ideal policy rate
- QE: used to try to influence long-term interest rates by purchasing assets

Balance sheets: CB balance sheet – Bank of England example

Figure 5.15 The Bank of England's (BoE) Weekly Report accounts for around 90% of the BoE's balance sheet. The data only extends backwards to September 2013. Before September 2013, we use the Bank Return for assets (which existed until September 2014, allowing us to verify overlap). Several asset series can be matched pre- and post-September 2013, but this is imperfect. Not all OMO series were reported before December 2008. We constructed post-September 2013 'other assets' to match the pre-September 2013 'other assets' as closely as possible. Prior to September 2013, foreign currency assets were counted under 'other assets,' and it is not possible to separate these. We assumed that the post-September 2013 series for bond holding assets would have been 60% of the pre-September 2013 equivalent when extending backwards, based on the overlap period.

Source: Bank of England's Weekly Report (data accessed June 2022).



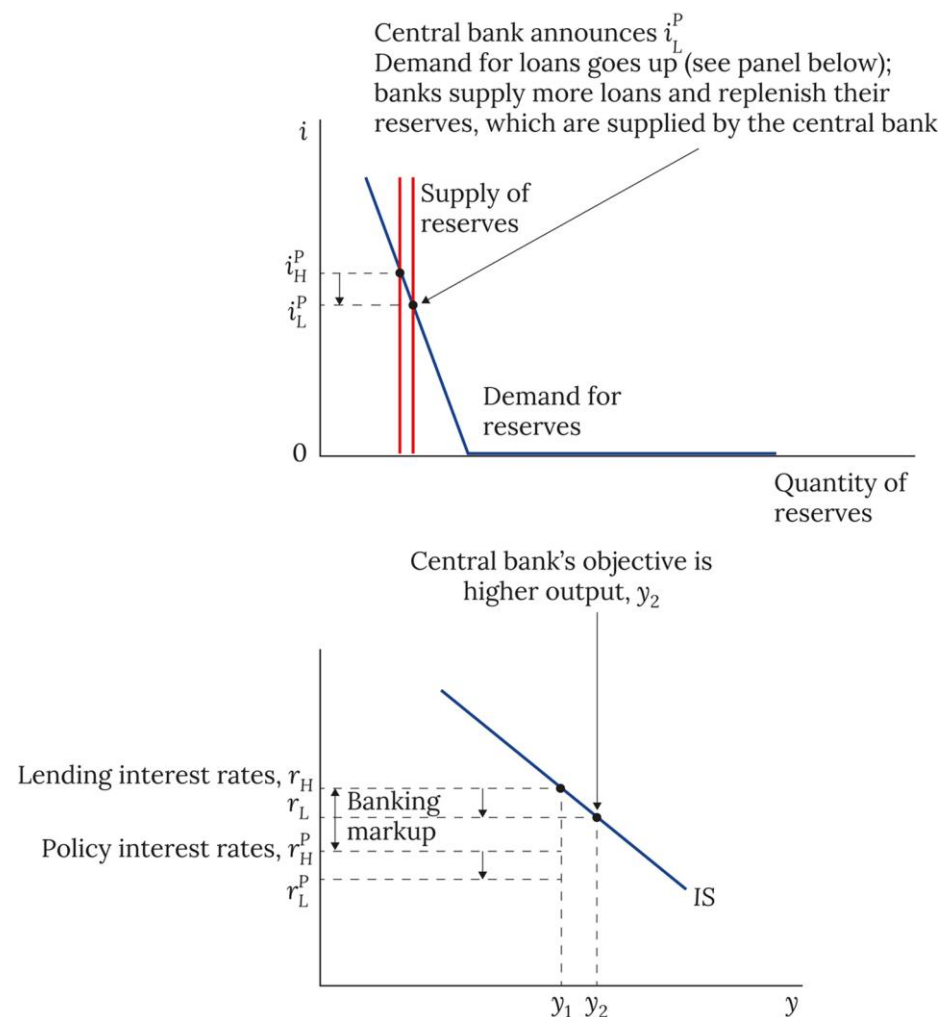
Note: Asset Purchase Facility (QE) is included under “other assets”

Pre-crisis: stable balance sheet, reflecting the success of inflation targeting through the setting of the policy rate (scarce reserves framework)

Post-crisis: huge growth in “other assets” driven by QE (ample reserves framework)

- “excess reserves”: money in reserve accounts that are over and above any reserves held for prudential reasons

Monetary policy operations: scarce reserves regime



Case: CB decides to raise its output target from y_1 to y_2

- IS Curve: as discussed before, CB targets r_L by announcing the policy rate r_L^P
- Market for reserves: downward sloping
 - ↓ lending rate after CB's announcement → ↑ loan demands → ↑ deposits → ↑ commercial banks' demand for reserves (liquidity purposes) → ↑ CB supplied reserves
- Hence, a fall of the policy rate is associated with an increase in reserves supply

Figure 5.16 The scarce reserves regime. The central bank implements a cut in the policy rate. Visit www.oup.com/he/carlin-soskice to explore the scarce reserves regime with Animated Analytical Diagram 5.16.

Monetary policy operations: scarce reserves regime

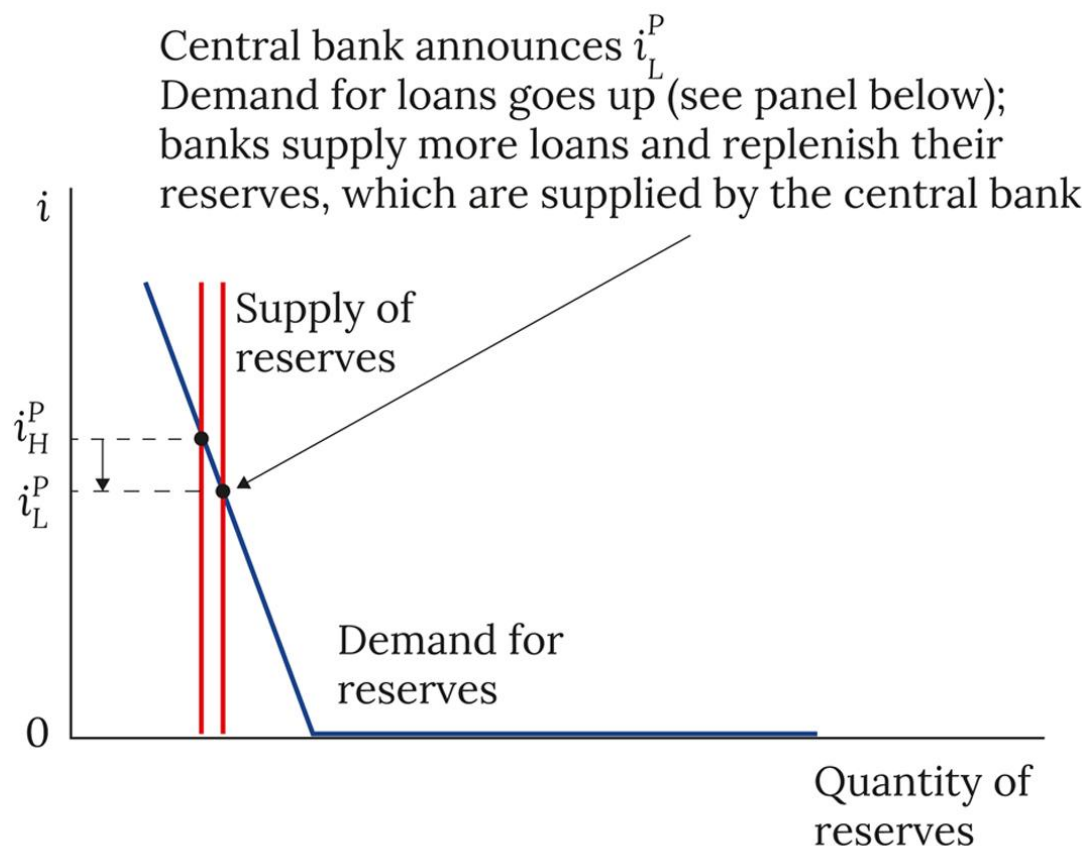


Figure 5.16 (top panel) The scarce reserves regime. The central bank implements a cut in the policy rate. Visit www.oup.com/he/carlin-soskice to explore the scarce reserves regime with Animated Analytical Diagram 5.16.

To increase supply of reserves, CB applies a combination of OMO and repos:

- E.g. performing its repo lending on a larger scale can increase the supply of reserves

Features of market for reserves:

- Demand for reserves is very inelastic, hence fluctuations in the supply of reserves will be modest
- Demand for reserves is flat when we hit the zero lower bound (no opportunity cost of holding reserves)

Monetary policy operations: ample reserves regime

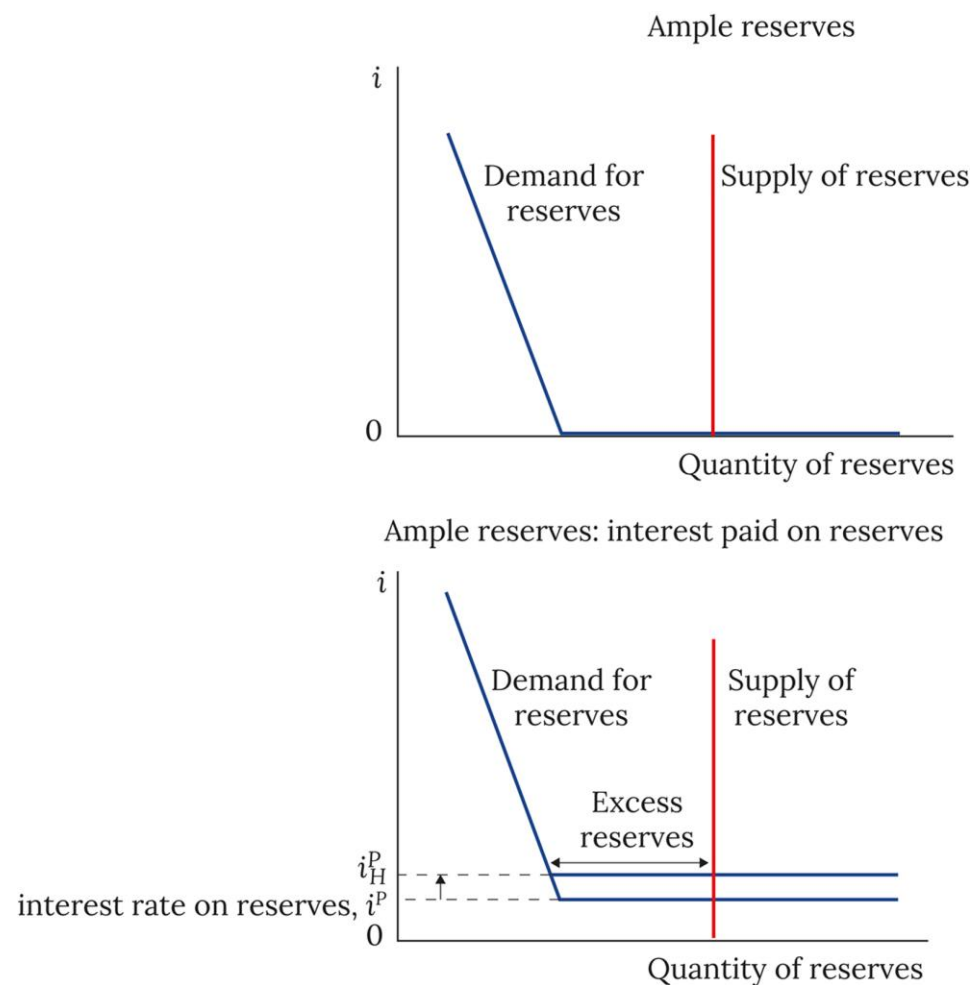


Figure 5.17 The ample reserves regime. The central bank implements a rise in the interest rate on reserves (lower panel).

Under ample reserves regime and zero interest on reserves, CB can only tighten monetary policy by unwinding its QE policy

- Given the large scale of QE, disruption in the financial market would be expected

Hence, CB chooses to pay interest on reserves under ample reserves

- The interest on reserves provides a floor to the “willingness to accept” price
- The demand curve is horizontal at this rate
 - No opportunity cost of reserves

Monetary policy operations: ample reserves regime

Ample reserves: interest paid on reserves

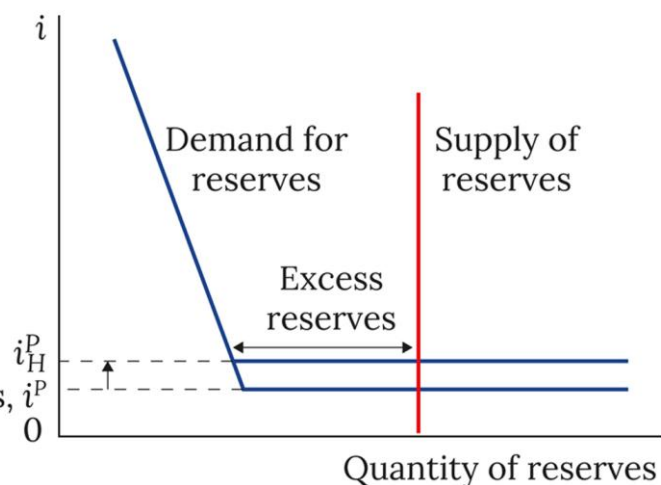


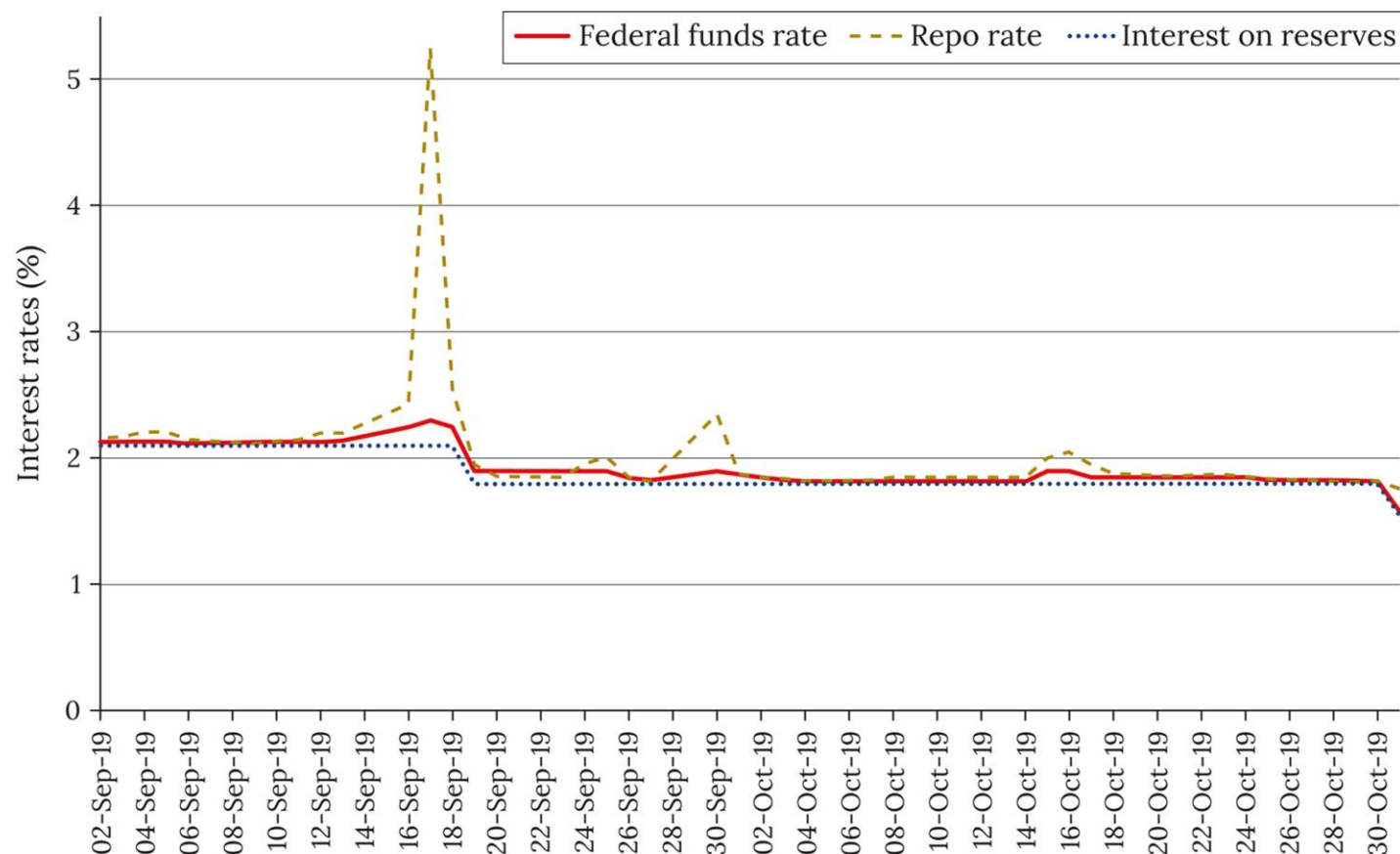
Figure 5.17 The ample reserves regime. The central bank implements a rise in the interest rate on reserves (lower panel).

CB can tighten monetary policy in the ample reserves regime with two choices: raise interest rate on reserves or quantitative tightening (QT):

- Raise interest rate on reserves:
 - $\uparrow$ interest rate on reserves $\rightarrow \uparrow$ money market rate $\rightarrow \uparrow$ lending rate $\rightarrow \downarrow$ demand for loans from the private sector
- QT: sell some of the bonds purchased during QE
 - CB sells bonds to primary dealers $\rightarrow$ dealers' banks settle transactions by decreasing dealers' deposits and transferring reserves to CB of the same amount $\rightarrow$ supply of reserves shrinks

Under the first option, the supply of reserves does not need to change, under the second, it must fall

Monetary policy operations: ample reserves regime - example



Due to uncertainty over the shape of the demand curve for reserves, QT should be applied gradually to avoid moving too rapidly into scarce reserves territory and losing control of the interest rate

- E.g., US in September 2019

Figure 5.18 United States interest rates in September and October 2019. We have used the effective federal funds rate as the money market (interbank) rate, the interest on excess reserves as the policy rate, and the Secured Overnight Financing Rate (SOFR) as a measure of the repo rate.

Source: Federal Reserve Bank of St. Louis (data accessed August 2021).

Summary:

The financial system serves many important functions:

- Facilitates economic transactions, allow maturity transformation, and enables inflation-targeting monetary policy

Balance sheets provide information on liabilities and assets of banks:

- They reflect liquidity and solvency of the banks, and help illuminate the relationship between the central bank and commercial banks

Building on the core 3-equation model, we have now included the banking system:

- The lending rate is a markup over the policy rate, as chosen by profit-maximizing banks

Summary:

The updated 3-equation model provides a valuable framework to study the banking system:

- The banking markup (r over r^P) is influenced by loan riskiness, risk-tolerance and the bank's capital cushion
- The CB reacts to economic shocks in a way similar as before (Ch. 3), except that now the CB sets the policy rate to target the lending rate that achieves the ideal output gap
- The application of QE changes the monetary policy framework from a scarce to an ample reserves regime
 - Interest rate on reserves is now the policy interest rate